

Intelligent Property Price Prediction and Agentic Advisory System

A Machine Learning Approach to Solving Real Estate Information Asymmetry

Why this industry matters: **context**, the **core problem**, and the **objective**

The Largest Global Asset Class

\$300T+

Real estate is the world's most significant store of wealth. Despite its massive scale, it remains highly illiquid and opaque compared to modern public equities markets.

Information Asymmetry

Buyers, sellers, and agents lack objective valuation mechanisms.

The market relies on biased, manual, and highly inconsistent appraisal methods—such as subjective broker estimates and slow comparable sales analysis.

Data-Driven Automation

The objective: Ingest structured property feature data to predict USD sale prices with high accuracy.

Output reliable confidence ranges to eliminate human bias from property valuation.

From Manual to Interactive

Transitioning the market from manual appraisals to an automated, interactive web UI—establishing the foundational ML engine for a future Agentic AI advisory system.

Model Inputs & Prediction Target / Output:

Inputs / Data Engineering

Inputs: 28 Distinct Features

Numeric (25):

- Area: Gr Liv Area, Lot Area, etc.
- Quality: Overall Qual, Overall Cond
- Age: Year Built, House Age
- Rooms: TotRms AbvGrd, Baths
- Garage Capacity & Efficiency

Categorical (3):

- Neighborhood, BldgType, House Style

Prediction Target & Output

The system yields a single predicted sale price in USD. It pairs this absolute value with:

- A confidence percentage
- An estimated error range ($\pm$ \$)
- Automated market comparable analysis

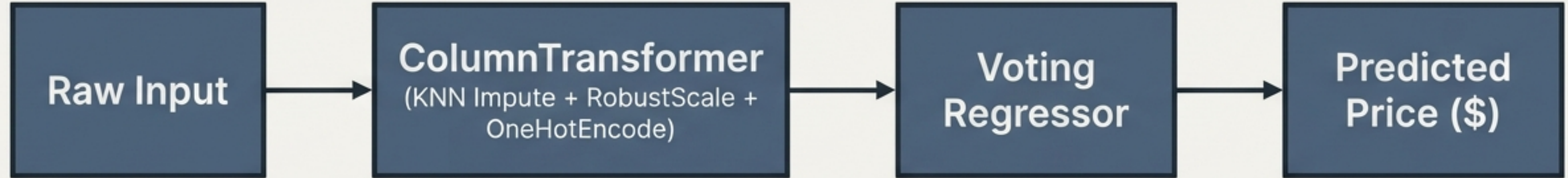
Data Engineering: Cleaning & Scaling

- Missing Values: KNN Imputation ($k=5$) preserves feature relationships.
- Outliers: IQR-based filtering ($1.5 \times$ IQR rule) prevents model skew.
- Scaling: RobustScaler applied to numeric features to handle remaining extremes.

Data Engineering: Feature Creation

- Generated 14 new features to capture market reality.
- Interaction terms: Quality $\times$ Area.
- Polynomials: House Age Squared, Log transforms.
- Encoding: OneHotEncoder ($\text{min_frequency}=2$) for categorical columns.

ML Architecture & Design / Pipeline Flow: Why Random Forest?



ML Architecture: Ensemble Components

- Random Forest: 300 trees, max_depth=22
- Gradient Boosting: 300 estimators, lr=0.04, depth=5
- Ridge Regression: alpha=0.1
- AdaBoost: 100 estimators, lr=0.05

Why Random Forest?

- **Non-linearity:** Effectively captures complex pricing interactions (e.g., quality compounds area value).
- **Outlier Robustness:** Voting across 300 trees ensures no single anomaly dominates the prediction.
- **Leakage Prevention:** Strict Scikit-Learn Pipeline integration ensures preprocessing steps fit only on training data.

Performance & Validation / Metrics Achieved: Interpretation

Metrics Achieved: Train vs. Test

Metric	Training	Testing
Accuracy	94.36%	90.81%
Precision	96.26%	90.24%
R ² Score	0.9626	0.9024
MAE	\$8,624	\$13,614
RMSE	\$11,307	\$19,055
MAPE	5.64%	9.19%

Performance & Validation: 5-Fold CV

Mean R² =

0.8789 ± 0.0298

The highly constrained standard deviation proves algorithmic stability across different data splits, verifying the absence of severe overfitting to any single fold.

Interpretation: Margin Analysis

43.11% of predictions land within exactly 5% of the actual transaction price.

72.99% of predictions land within 10% of the actual transaction price.

Interpretation: Controlled Variance

The ~4% performance gap between training accuracy (94.36%) and test accuracy (90.81%) represents a controlled, acceptable variance inherent to real estate anomalies, rather than a systemic modeling failure.

Deployment & Real-World Impact

Application Features: Real-World Benefits

Application Features: Interface

- **Property Input Form:** Intuitive Streamlit UI with dropdowns and sliders for area, quality ratings, rooms, and neighborhood.
- **Instant Predictions:** Real-time USD output generated alongside transparent confidence ranges and error estimates.

Application Features: Analytics

- **Market Dashboard:** Integrated visualizations for market trends, feature importance mapping, and neighborhood price distribution.
- **Model Persistence:** The trained ensemble is saved via `joblib` to guarantee fast, continuous inference without re-training delays.

Real-World Benefits

- **Homebuyers:** Sanity-check listing prices against algorithmic realities prior to negotiation.
- **Real Estate Agents:** Generate immediate, objective automated valuations for clients.
- **Institutional Investors:** Screen entire portfolios against predicted vs. listed prices to rapidly identify undervalued assets.

Milestone 2: Agentic AI Advisory

The current predictive engine is the foundation for an upcoming autonomous advisory system.

Utilizing `LangGraph`, `Mistral/Llama2`, and `Chroma RAG`, the next phase will transition the tool from passive price prediction to active, autonomous investment report generation.